Monthly Intelligence Report

Edition 001 — The Puglia Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

4,293

475 HV · 3,818 MV

GRID LINES MAPPED

14,221

~46,000 km coverage

MC SIMULATIONS

42.9 M

10,000 per substation

PUBLIC DATA SOURCES

30

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 4,293 substations and 14,221 grid lines spanning ~46,000 km.

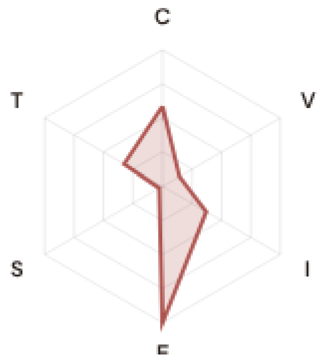
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Italy's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Magisano CP

Cosenza, Calabria · 150 kV

0.547	High	0.212	86th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

C Continuity: 0.585 / 0.30 (195%)

I Infrastructure: 0.370 / 0.25 (148%)

S Saturation: 0.026 / 0.20 (13%)

V Voltage: 0.142 / 0.10 (142%)

E Economic: 1.000 / 0.10 (1000%) ⚠

T Transition: 0.324 / 0.05 (648%)

MODIFIERS

R3 Consequence: 1.134 ↑ amplifies

R4 Graph Criticality: 1.050 ↑ amplifies

R6a Restoration: 0.979 ↓ dampens

R7 Digital Readiness: 0.986 ↓ dampens

This month's spotlight — automatically selected for April 2026 — is **Magisano CP** in Cosenza, Calabria. With an R_final of 0.547 (High band, 86th fleet percentile), its risk profile is dominated by the E (Economic) component at 1000% of its weight ceiling, followed by T (Transition) at 648%. Modifiers collectively shift R_base by 35.4%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that PNRR digitalisation planning requires.

CYBER HIGH EXPOSURE

669

15.6% of fleet

BLIND SPOTS

872

High/Critical + R7 < median

AVG R7 MODIFIER

1.0009

Range [0.99, 1.05]

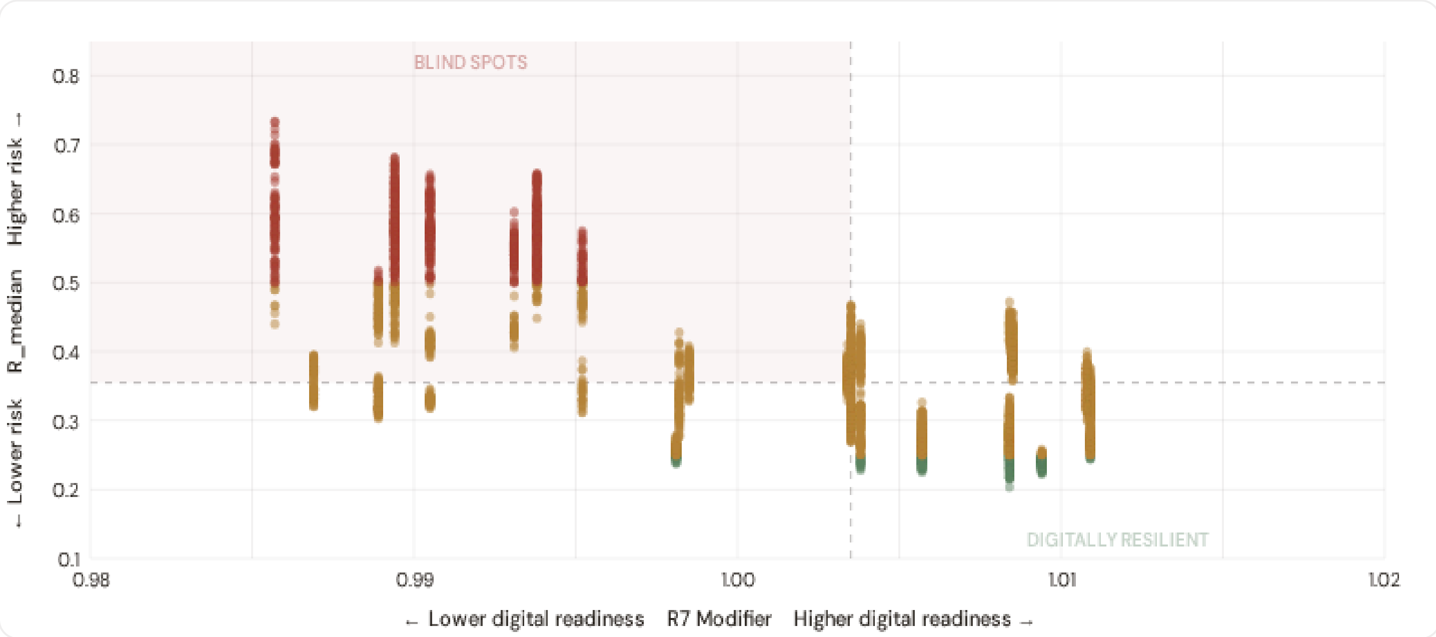
HIGH-CONSEQUENCE SUBS

3,611

84.1% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **872 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0035). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Sicilia (240), Campania (213), Puglia (152)**. Across the full fleet, 669 substations (15.6%) carry a HIGH cyber-exposure classification, while 207 (4.8%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in the northern urban grid.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: €15–30/kWh

Heavy industry, petrochemical, large manufacturing — highest economic loss per interruption

3,179 substations (74.1%) High/Critical: **320** (10.1%) Avg R: **0.355** Avg E: **0.407** / 0.10

Low 167 Med 2692 High 320 Crit 0

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: €8–15/kWh

Diversified industry, logistics, food processing — significant continuity requirements

740 substations (17.2%) High/Critical: **339** (45.8%) Avg R: **0.473** Avg E: **0.685** / 0.10

Low 22 Med 379 High 339 Crit 0

🏪 Commercial / SME-Dense

Est. VoLL: €3–8/kWh

Service economy, retail, tourism — moderate individual impact but high aggregate exposure

231 substations (5.4%) High/Critical: **213** (92.2%) Avg R: **0.565** Avg E: **0.868** / 0.10

Low 0 Med 18 High 213 Crit 0

🌾 Light / Agricultural / Rural

Est. VoLL: €1–3/kWh

Sparse economic activity, agricultural land — lower VoLL but higher social vulnerability

143 substations (3.3%) High/Critical: **0** (0.0%) Avg R: **0.241** Avg E: **0.035** / 0.10

Low 131 Med 12 High 0 Crit 0

Value of Lost Load (VoLL) context: ACER's 2023 estimate for Italy places average VoLL at €11.4/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **320 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Lombardia (632), Toscana (319), Emilia-Romagna (298)** — regions where industrial corridors depend on uninterrupted power for continuous processes (steel, glass, automotive). A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 10.0%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify provinces combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 PROVINCES

Longer bar = higher R7 = better digital infrastructure.

1	Vibo Valentia ⚠ high R	0.9857
2	Cosenza ⚠ high R	0.9857
3	Reggio Calabria ⚠ high R	0.9857
4	Crotone ⚠ high R	0.9857
5	Catanzaro ⚠ high R	0.9857
6	Matera	0.9869
7	Potenza ⚠ high R	0.9869
8	Isernia	0.9889
9	Campobasso ⚠ high R	0.9889
10	Messina ⚠ high R	0.9894
11	Agrigento ⚠ high R	0.9894
12	Enna ⚠ high R	0.9894
13	Palermo ⚠ high R	0.9894
14	Catania ⚠ high R	0.9894
15	Ragusa ⚠ high R	0.9894

Province-level analysis reveals a clear North-South digital divide mirroring the broader DESI pattern. **Vibo Valentia** has the lowest average R7 (0.9857), while **Parma** leads at 1.0109 — a spread of 0.0252 across the R7 range. **44 provinces** combine below-median digital readiness with above-median grid risk, making them priority targets for PNRR digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0009, median = 1.0035; 669 substations classified HIGH cyber-exposure; 872 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging provinces where DESI improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **35 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL

35
95 variables

5
High-frequency feeds

22
Regular refresh cycle

STATIC / DERIVED
8
Standards + scenarios

Data Source Registry — April 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
TERNA-N	Terna Piano di Sviluppo	MONTHLY	TSO zone	2 vars
RSE	RSE (Ricerca sul Sistema Energetico)	CONTINUOUS	Provincia	1 var
GEOPNAT	Geoportale Nazionale (TERNA / GSE Atlaimpianti)	QUARTERLY	Commune	5 vars
GSE	GSE (Gestore dei Servizi Energetici)	QUARTERLY	Provincia	1 var
ENEA-INN	ENEA Innovation / RSE	QUARTERLY	Provincia	1 var
ARERA	ARERA (Autorità de Régulation de l'Énergie)	ANNUAL	Provincia	8 vars
ISTAT	ISTAT (Istituto Nazionale di Statistica)	ANNUAL	Provincia	8 vars
ENEA	ENEA (Agenzia Nazionale Nuove Tecnologie Energia)	ANNUAL	Provincia	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
MASE	MASE (Ministero Ambiente e Sicurezza Energetica)	ANNUAL	Région	2 vars
ISPRA-AMB	Cerema (Centre d'études risques)	ANNUAL	Provincia	1 var
MEF	MEF / Agenzia delle Entrate	ANNUAL	Provincia	1 var
INGV	INGV (Istituto Nazionale Geofisica e Vulcanologia)	ANNUAL	Provincia	1 var
ISTAT-POV	ISTAT Poverty / BES Observatory	ANNUAL	Provincia	2 vars
EDIST	e-distribuzione Open Data	ANNUAL	Provincia	2 vars
ANCI	ANCI Observatoire des Territoires	ANNUAL	Provincia	2 vars
ARERA-M	ARERA Monitoring Report	ANNUAL	DSO-level	2 vars
CGET	CGET/ANCI Spatial Planning	ANNUAL	Provincia	1 var
SVIMEZ	SVIMEZ Economic Research	ANNUAL	Région	1 var
ENEA-R	ENEA Regioni	ANNUAL	Région	1 var
ISPRA	ISPRA (Istituto Superiore Protezione e Ricerca Ambientale)	STATIC	Provincia	3 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars

DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
COG	COG ISTAT Provincia Registry	STATIC	Provincia	1 var
ISO-9223	ISO 9223 Corrosion	DERIVED	Provincia	1 var
TERNA	Terna Transparency (Gaudi)	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
MF-OM	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
D-OM	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources) — OSM diffs + weather reanalysis

OSM Overpass · Open-Meteo/ERA5

Monthly (3 sources) — Grid load + procurement data

GSE Atlaimpianti · Terna Gaudi · Terna Piano Sviluppo

Quarterly (2 sources) — DER registry + business data + macro indicators

GSE · Bpiitaly

Annual (20 sources) — Core grid quality + demographics + environment

ARERA · ISTAT · ISTAT-POV · e-distribuzione · MASE · ENEA · MEF · EEA · Eurostat · DESI · ANCI · ARERA Monitoring · CGET · SVIMEZ · ENEA-R · INGV · Open-Meteo · JRC · RSE · Cerema

Static (8 sources) — Standards + scenarios + seismic hazard

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · GFZ · ISPRA · ISO 9223 · COG/ISTAT · ENTSO-E

C.2 Fleet Score Snapshot

FLEET SIZE
4,293
substations scored

MEDIAN R
0.355
P5=0.244 · P95=0.603

HIGH + CRITICAL
872
20.3% of fleet

HIGH CONFIDENCE
100%

4292 subs · CI/R < 0.50

BAND DISTRIBUTION

No primary data sources have been updated since the v4.0.2 launch baseline. All **4,293 substation scores remain unchanged** this edition. The fleet median R of 0.355 (mean 0.383) reflects a distribution skewed toward the lower bands, with 7.5% in Low and 72.2% in Medium. The first material score changes are expected when ARERA publishes the annual TIQE tables (affecting C1–C4 and R6a) and when GSE refreshes the Atlaimpianti DER registry (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — Provincia-level DESI model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — HRST + startup density	§6
L3	ENHANCED	V_socio Fiscal Enrichment — MEF + ANCI + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — ISTAT net migration	§5
L5	DATA	95/95 variables operational (100%). 35 data sources total.	§8, §12
G1	DATA	d02 GSE Atlaimpianti upgraded to quarterly registry — Provincia-level DER registry	§12
G2	DATA	d05 ISPRA upgraded to live API — Provincia-level geological data	§12
G3	DATA	d06 OSM upgraded to Overpass API — 7,898 real substations	§12
G4	NEW	R6b Network Topology modifier — centrality + ring analysis from OSM graph	§5
G5	NEW	Network & Topology layer (H): 7 variables — Terna Piano di Sviluppo, ring analysis	§12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Puglia Corridor: Where the Energy Transition Is Outrunning the Grid

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Puglia corridor · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North-South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Provincial Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why Puglia, Why Now

PUGLIA SUBSTATIONS

238

147 HV · 91 MV

HIGH BAND

152

63.9% of corridor

CORRIDOR MEDIAN R

0.548

Fleet median: 0.355

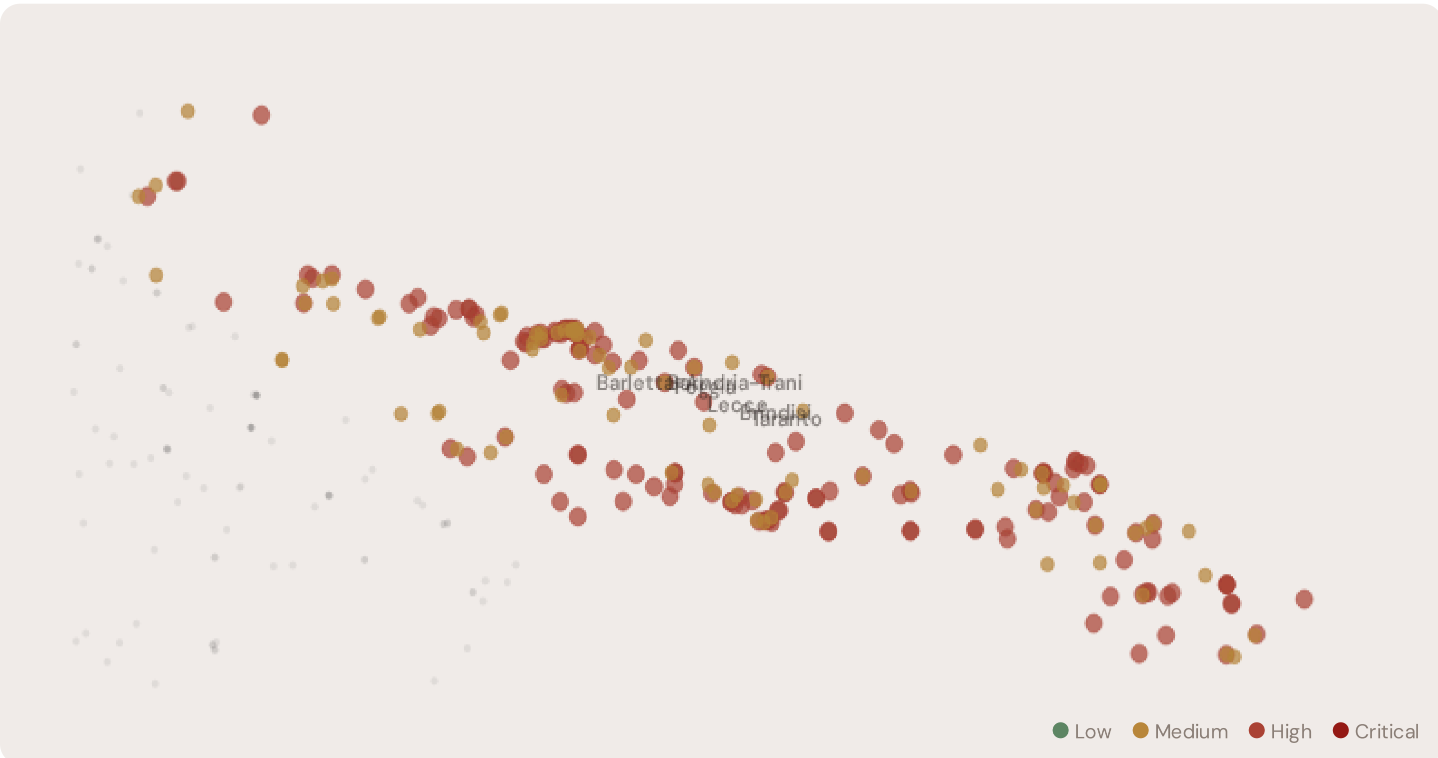
WORST PROVINCE

Brindisi

Avg R: 0.622 · 47 substations

Puglia hosts **238 substations** across 6 provinces, making it the 6th largest regional fleet in the SSI dataset. However, its risk profile is disproportionately concentrated in the upper bands: **152 substations (63.9%)** are classified High, with only **0** in the Low band. 84% of Puglia substations score above the national fleet median of 0.355. The corridor median R of 0.548 places Puglia among the most stressed regions in the Mezzogiorno, driven by the combination of high continuity-of-supply deficits, ageing infrastructure, and accelerating DER penetration under PNRR deployment.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Erchie	Brindisi	0.657	High	1.000	0.234	0.353	0.790	0.043	0.724	C, E, T
Taranto Smistamento	Brindisi	0.653	High	0.964	0.218	0.363	0.817	0.040	0.716	C, E, T
Firestone	Brindisi	0.650	High	0.940	0.259	0.345	0.806	0.042	0.772	C, E, T
Loseto	Brindisi	0.649	High	0.947	0.250	0.346	0.846	0.043	0.736	C, E, T
Carpignano Salentino	Brindisi	0.646	High	0.933	0.216	0.364	0.819	0.044	0.726	C, E, T
Turi	Brindisi	0.642	High	0.951	0.228	0.328	0.791	0.043	0.748	C, E, T
Acquaviva delle Fonti	Brindisi	0.636	High	0.884	0.240	0.362	0.834	0.039	0.692	C, E, T
Francavilla	Brindisi	0.635	High	0.917	0.236	0.349	0.803	0.042	0.742	C, E, T
Bari FS	Brindisi	0.633	High	0.922	0.236	0.334	0.863	0.043	0.676	C, E, T
Sub_24744	Brindisi	0.632	High	0.883	0.260	0.332	0.848	0.039	0.732	C, E, T

… 228 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — PUGLIA VS FLEET

C — Continuity 0.622 vs 0.362 (+72%)

I — Infrastructure 0.338 vs 0.394 (−14%)

S — Saturation 0.041 vs 0.066 (−38%)

V — Voltage 0.238 vs 0.151 (+58%)

E — Economic 0.819 vs 0.467 (+75%)

T — Transition 0.707 vs 0.458 (+54%)

MODIFIER AVERAGES — PUGLIA VS FLEET

R3 Consequence	1.145	fleet 1.137	↑
R4 Graph Crit.	1.050	fleet 1.055	↑
R6a Restoration	0.991	fleet 1.004	→
R7 Digital Read.	0.990	fleet 1.001	→

The dominant risk driver across the Puglia corridor is **T (Transition)**, averaging 0.707 against a fleet average of 0.458 — occupying 1413% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.819 vs fleet 0.467 (819% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.145 (fleet: 1.137), reflecting the socio-economic fragility of Puglia's service-oriented, tourism-dependent economy. The R6b seismic modifier averages 1.025, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

PROVINCE RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Brindisi 47 subs · avg R = 0.622 · H:47 M:0

Taranto 39 subs · avg R = 0.572 · H:39 M:0

Foggia 38 subs · avg R = 0.569 · H:38 M:0

Lecce 30 subs · avg R = 0.524 · H:28 M:2

Barletta-Andria-Trani 45 subs · avg R = 0.414 · H:0 M:45

Bari 39 subs · avg R = 0.331 · H:0 M:39

The province-level breakdown reveals significant intra-regional variation. **Brindisi** leads with a mean R of 0.622 across 47 substations, while **Bari** scores 0.331 — a spread of 0.291 within a single region. This disparity underscores why regional-level metrics (as used by CEER and JRC) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that Puglia is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

3 Puglia substations score R ≥ 0.650, placing them in the upper quartile of national risk. For PNRR investment prioritisation, this analysis identifies the Brindisi corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by service-economy dependence, tourism exposure, and above-average energy poverty — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

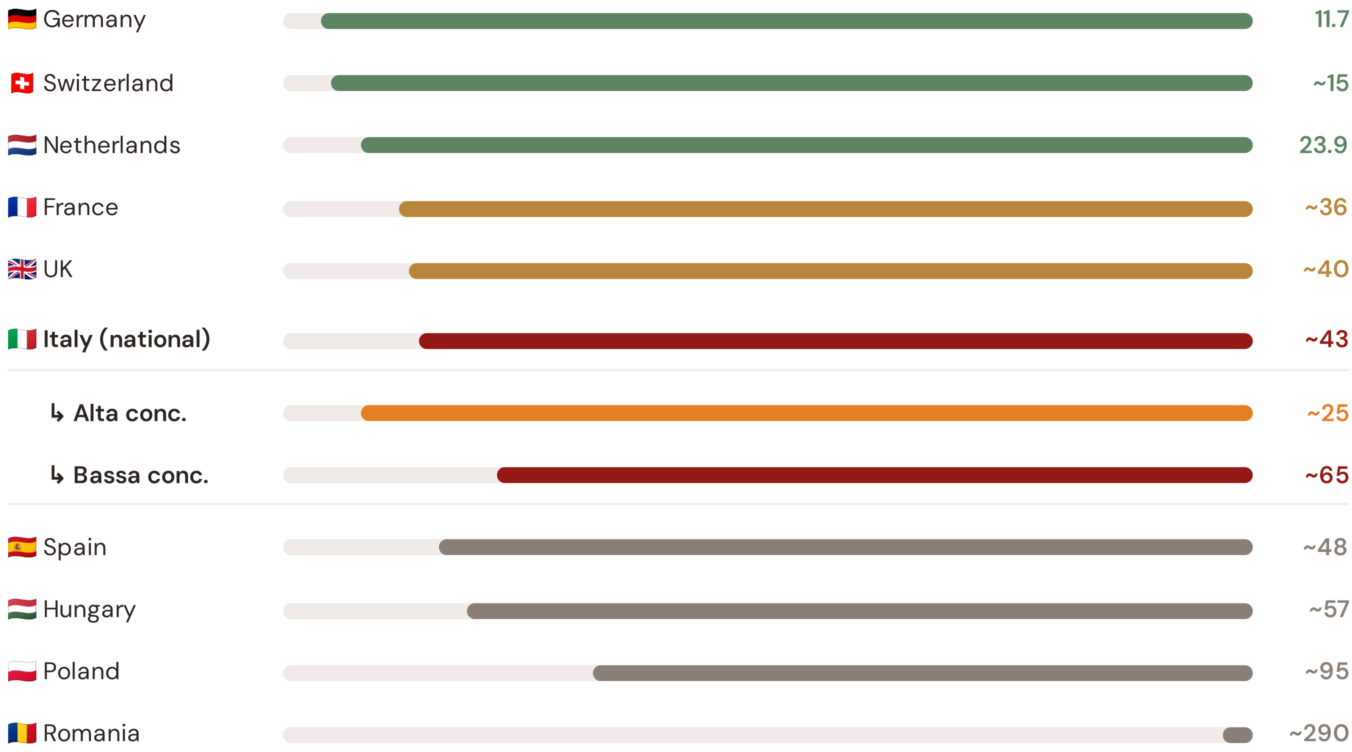
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Puglia corridor analysis hinges on Italy's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

EUROPEAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



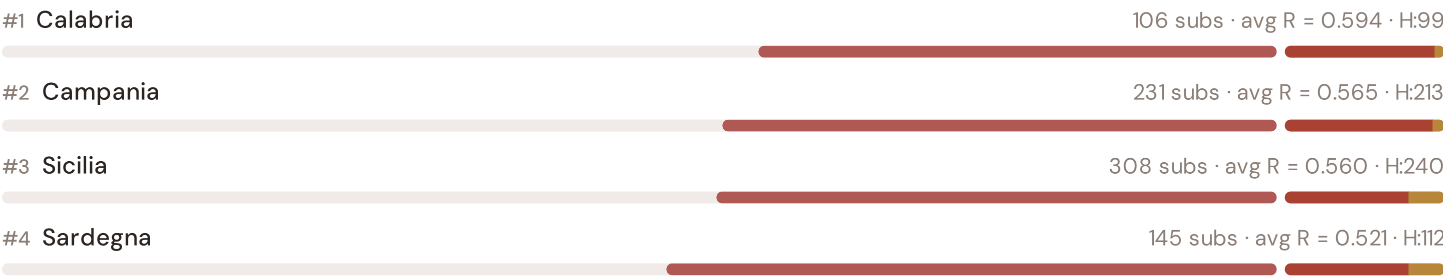
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). Italy sub-categories from ARERA territory classification. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

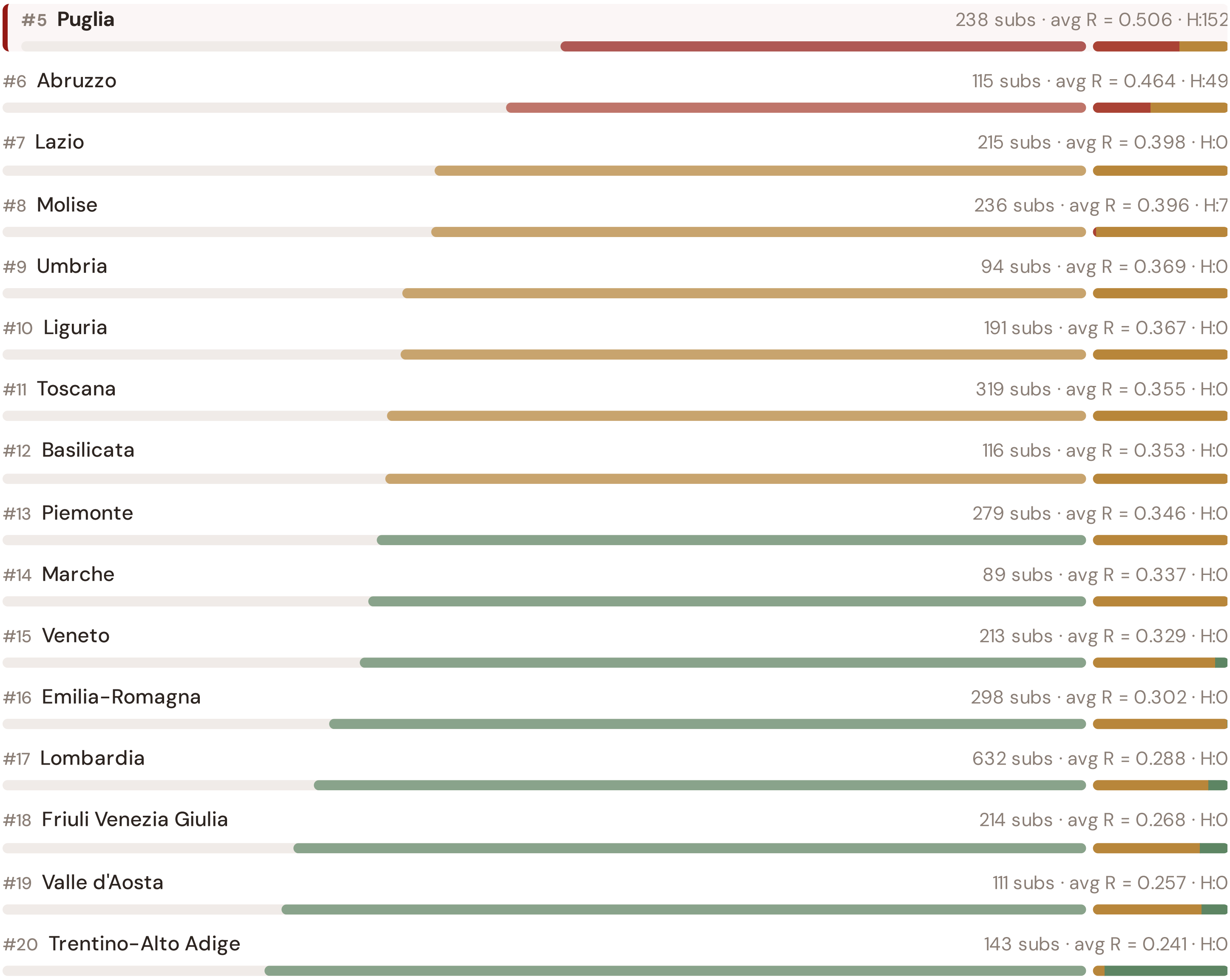
This month's key insight: The Puglia corridor deep-dive shows **208 substations** (of 238) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Italy's "bassa concentrazione" territory class. Puglia's average C of 0.622 is **1.7× the fleet average** of 0.362. In European terms, these substations individually perform closer to Hungarian or Spanish SAIDI levels (~48–57 min/customer/year) than to Italy's own national average (~43 min). The SSI reveals what aggregate CEER statistics conceal: that Italy's mid-tier European position is not a national condition but a statistical average of Dutch-quality urban performance and Eastern-European-level rural performance — and the Puglia corridor concentrates the worst of the latter.

Regional Risk Ranking — All 20 Italian Regions

Where does Puglia sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.





Puglia ranks **#5 of 20 regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed regions are Calabria, Campania, Sicilia, while the three most resilient are Trentino-Alto Adige, Valle d'Aosta, Friuli Venezia Giulia. 8 of 20 regions score above the fleet average of 0.383. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress across Italy. It is precisely this substation-level granularity that positions the SSI as a tool for PNRR investment prioritisation: not treating "Southern Italy" as a monolith, but identifying the specific corridors within each region that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DNO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering a €10M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history — and can verify this against ARERA's published TIQE data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall visualisation: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The Italian training set comprises 4,293 substations with complete 95-feature vectors. Average CI_width of 0.152 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 320 Low, 3101 Medium, 872 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

The Lazio Metropolitan Load

Deep-dive into Lazio's grid risk profile — substation map, component analysis, province breakdown. European Context Card: DER + EV penetration stress on the Roma metropolitan grid.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

GSE, Bpiitaly are due for refresh in Q3. Impact: DER registry updates (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **ARERA (Autorità de Régulation de l'Énergie)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of May 2026. Deep-dive region: **Lazio**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Italian utility, ARERA, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 4,293 substations · 14,221 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5